ENVS 193DS Homework 03

Elena Donelson

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Setup

```
# loading in packages
library(tidyverse)
library(janitor)
library(gt)
library(rstatix)

# reading in kelp data
kelp <- read_csv("temp-kelp.csv")

# reading in my coffee data
coffee_data <- read_csv("coffee_data 2.csv")
```

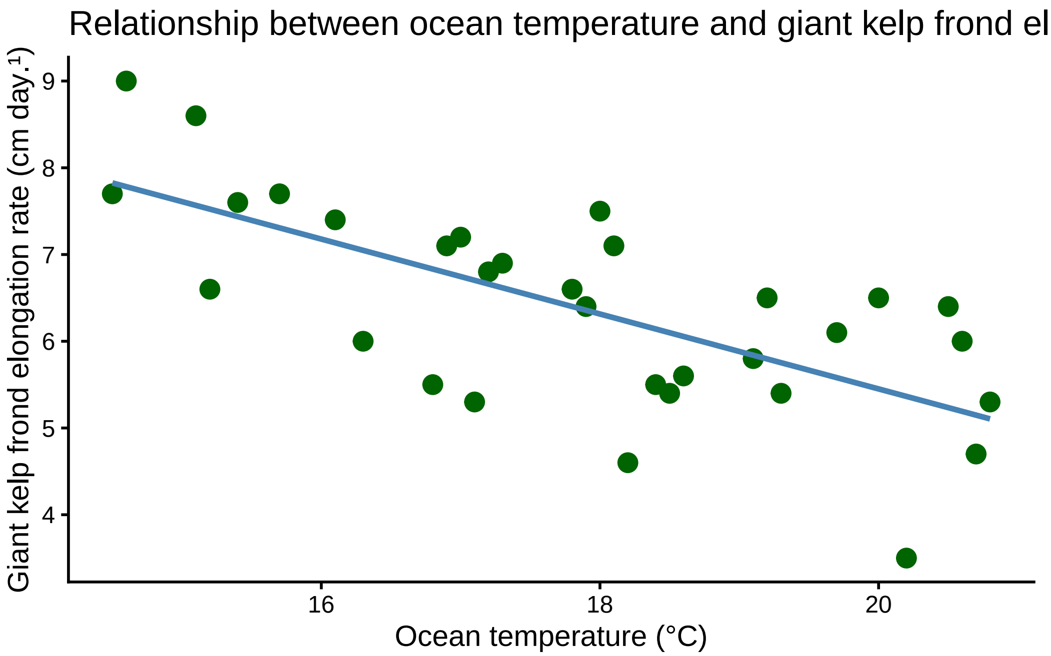
Problem 1. Giant kelp fronds

a.

You could use either a Pearson correlation test or a simple linear regression to determine the strength of the relationship between ocean temperature (°C) and giant kelp frond elongation rate (cm day ^{1}). Both tests analyze the relationship between two continuous quantitative variables, but a Pearson correlation measures the strength and direction of the relationship, while a linear regression tests how changes in ocean temperature predict changes in giant kelp frond elongation rate.

b.

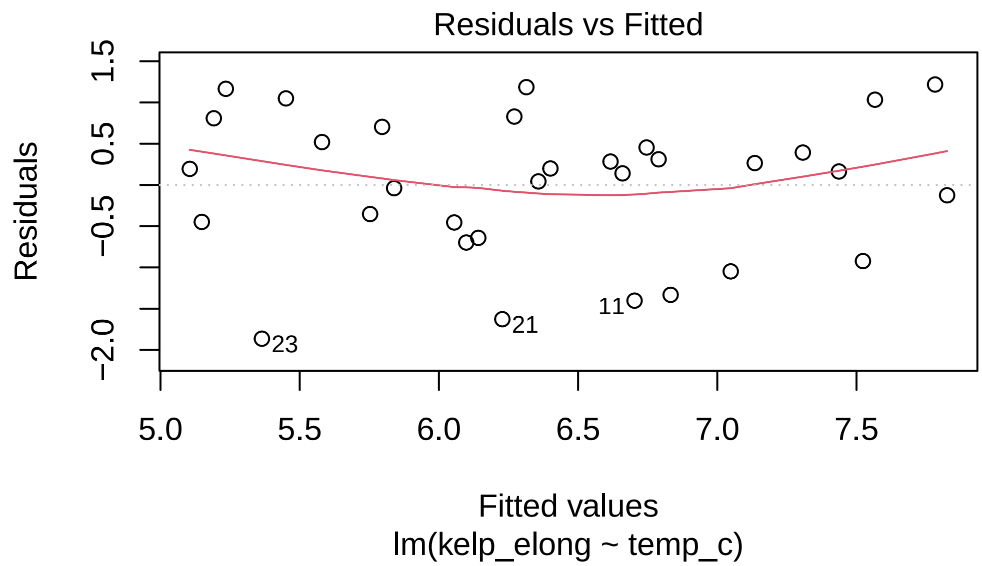
```
# creating a scatter plot showing the relationship between ocean temperature
# and giant kelp frond elongation rate
ggplot(
  kelp,
  aes(
    x = temp_c,
    y = kelp_elong
  )
) +

  # plotting individual data points
  geom_point(
    color = "darkgreen",
    size = 3
  ) +

  # adding a trend line
  geom_smooth(
    method = "lm",
    se = FALSE,
    color = "steelblue"
  ) +

  # changing the theme
  theme_classic() +

  # labeling the axes and title
  labs(
    title = "Relationship between ocean temperature and giant kelp frond elongation rate",
    x = "Ocean temperature (°C)",
    y = "Giant kelp frond elongation rate (cm day ¹)"
  )
```

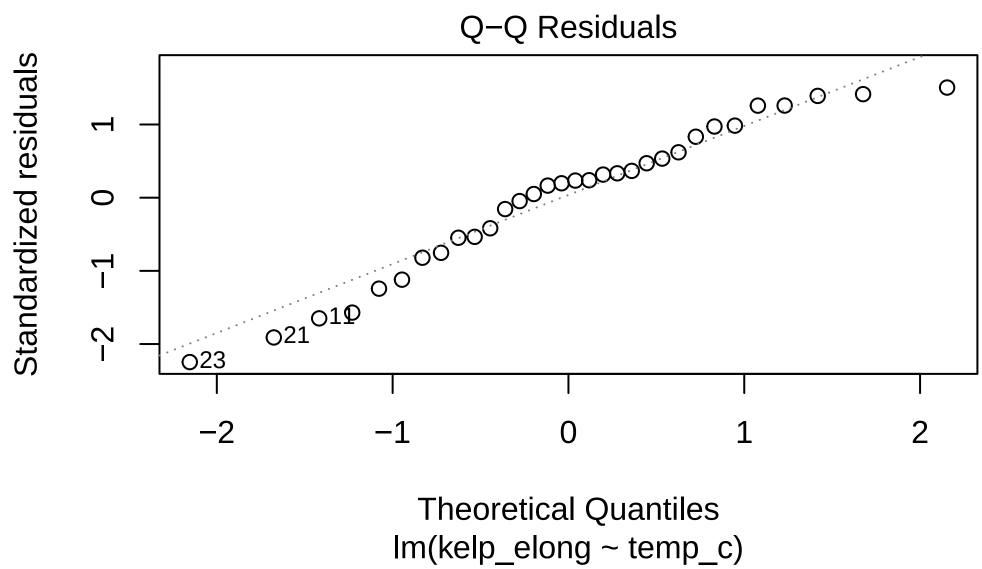


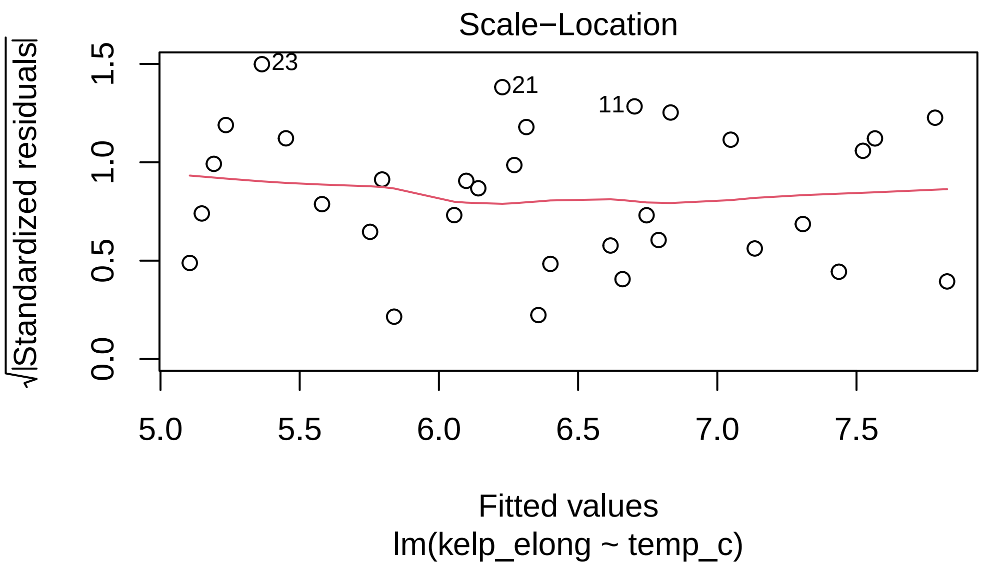
c.

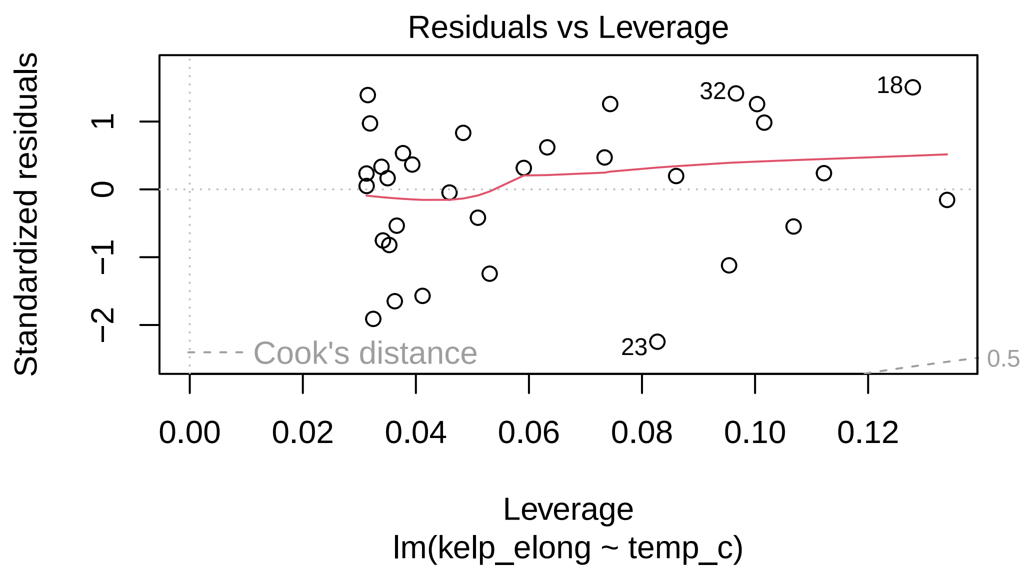
Checking assumptions

```
# creating a linear model to test the relationship between ocean temperature
# and giant kelp frond elongation rate
kelp_lm <- lm(
  kelp_elong ~ temp_c,
  data = kelp
)

# creating diagnostic plots to check linear regression assumptions
plot(kelp_lm)
```


Residuals vs Fitted
Residuals
1.5
0.5
−0.5
−2.0
5.0
5.5
6.0
6.5
7.0
7.5
23
21
11
Fitted values
lm(kelp_elong ~ temp_c)


Q−Q Residuals
Standardized residuals
1
0
−1
−2
−2
−1
0
1
2
23
21
11
Theoretical Quantiles
lm(kelp_elong ~ temp_c)





I checked the assumptions for a simple linear regression, including linearity, normality of residuals, and equal variance. I checked these assumptions using the diagnostic plots from the linear model. Based on the plots, the assumptions appear reasonable enough to continue with the linear regression test.

Running the test

```r
# running a simple linear regression to test whether ocean temperature
# predicts giant kelp frond elongation rate
summary(kelp_lm)
```

```
Call:
lm(formula = kelp_elong ~ temp_c, data = kelp)

Residuals:
    Min      1Q  Median      3Q     Max 
-1.8643 -0.5017  0.1790  0.5659  1.2177 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 14.08645    1.49219   9.440 1.72e-10 ***
temp_c      -0.43179    0.08321  -5.189 1.37e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom
Multiple R-squared:  0.473,	Adjusted R-squared:  0.4554 
F-statistic: 26.93 on 1 and 30 DF,  p-value: 1.366e-05
```

A simple linear regression showed a significant relationship between ocean temperature and giant kelp frond elongation rate ($F = 26.93$, $df = 1, 30$, $p < 0.001$). Temperature was a significant predictor of elongation rate ($= -0.432$, $p < 0.001$), indicating that kelp frond elongation rate decreased as ocean temperature increased. The model explained approximately 47.3% of the variation in elongation rate ($R^2 = 0.473$).

d.

To evaluate the relationship between ocean temperature and giant kelp frond elongation rate, I used a simple linear regression. I used this test because both variables are continuous quantitative variables, and I wanted to see whether ocean temperature could predict changes in giant kelp frond elongation rate.

The linear regression showed a significant negative relationship between ocean temperature and giant kelp frond elongation rate ($F = 26.93$, $df = 1, 30$, $p < 0.001$, $R^2 = 0.473$). As ocean temperature increased, giant kelp frond elongation rate decreased, suggesting that warmer ocean temperatures are associated with slower giant kelp growth.

e.

The results suggest that higher ocean temperatures are associated with slower giant kelp growth. This could mean that continued ocean warming may negatively affect giant kelp forests by reducing their growth rates and overall health. Since giant kelp provides important habitat for many marine species, monitoring ocean temperatures and kelp growth could help identify areas that may be more vulnerable to climate change.

f.

```
# running a Pearson correlation test to determine the strength and direction
# of the relationship between ocean temperature and giant kelp frond elongation rate
cor.test(
  kelp$temp_c,
  kelp$kelp_elong,
  method = "pearson"
)
```

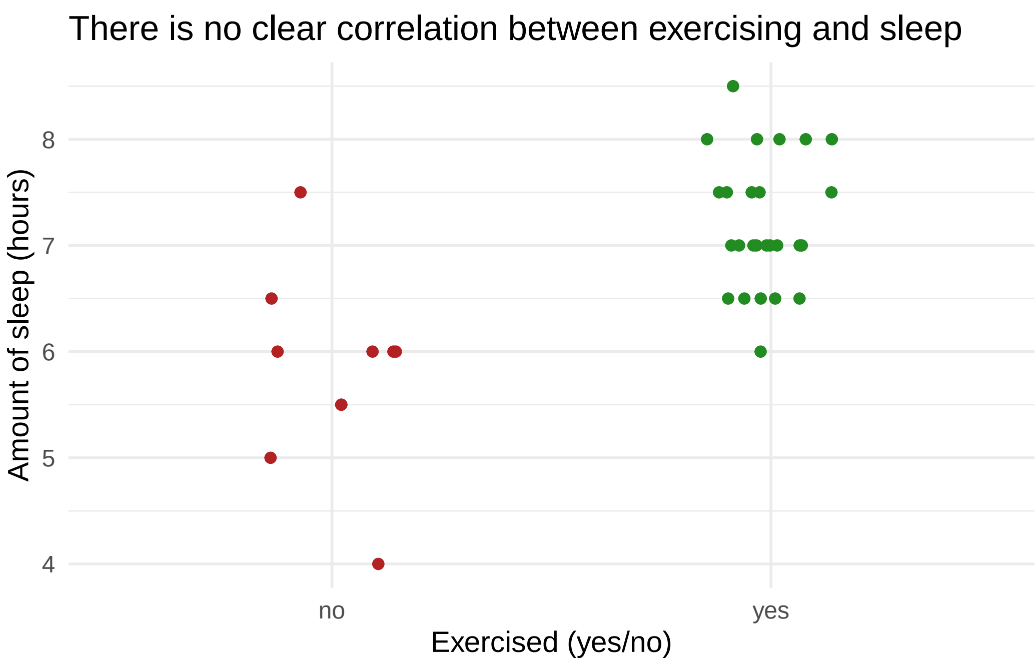
```
	Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
       cor
-0.6877489
```

The Pearson correlation test and the simple linear regression would have led me to the same conclusion because both tests found a significant relationship between ocean temperature and giant kelp frond elongation rate ($p < 0.001$). The Pearson correlation showed a moderately strong negative relationship between the two continuous variables ($r = -0.688$), while the linear regression showed that giant kelp frond elongation rate decreases as ocean temperature increases. Therefore, both tests indicate that warmer ocean temperatures are associated with slower giant kelp growth.

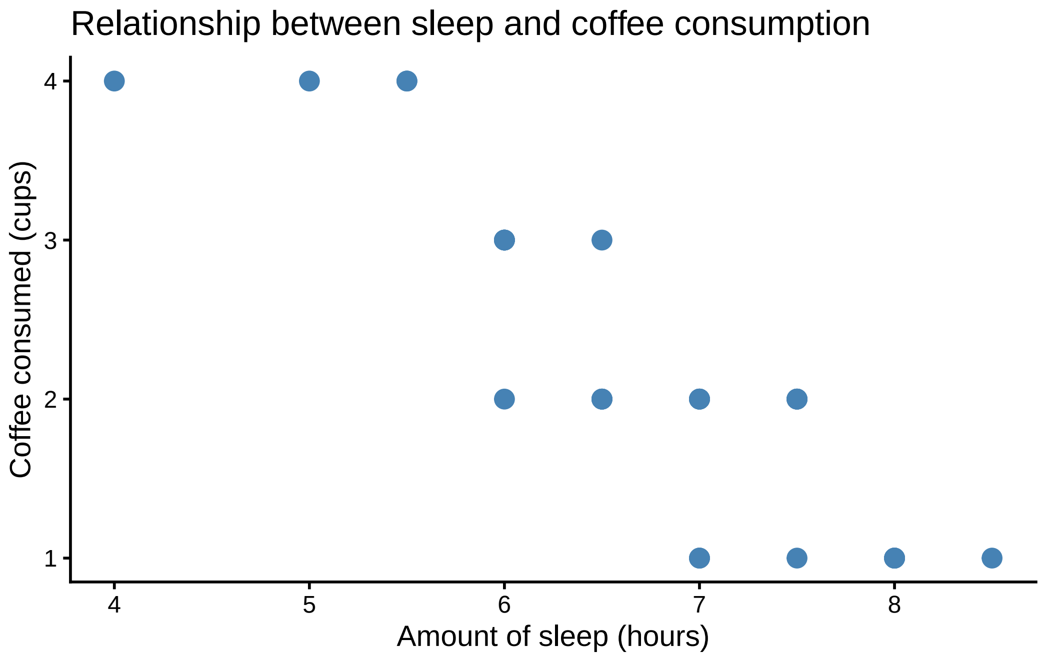
1. Problem 2. Personal data

a.

```r
# creating a plot that shows the relationship between exercise and hours slept
ggplot(
  coffee_data,
  aes(
    x = exercise,
    y = hours_slept,
    color = exercise
  )
) +
  # plotting individual data points
  geom_jitter(width = 0.15, height = 0) +

  # setting custom colors for groups
  scale_color_manual(values = c(
    "yes" = "forestgreen",
    "no" = "firebrick"
  )) +

  # changing the theme
  theme_minimal() +

  # adding a title and labeling the x and y-axis
  labs(
    title = "There is no clear correlation between exercising and sleep",
    x = "Exercised (yes/no)",
    y = "Amount of sleep (hours)"
  ) +

  # removing the legend
  theme(legend.position = "none")
```




```r
# creating a scatter plot of sleep data and coffee consumption
ggplot(
  coffee_data,
  aes(
    x = hours_slept,
    y = coffee_cups
  )
) +
  # plotting individual data points
  geom_point(color = "steelblue", size = 3) +

  # changing the theme
  theme_classic() +

  # adding a title and naming the x and y-axis
  labs(
    title = "Relationship between sleep and coffee consumption",
    x = "Amount of sleep (hours)",
    y = "Coffee consumed (cups)"
  )
```



b.

Figure 1. Relationship between exercise status and amount of sleep (hours) from April–May 2026. Points represent individual daily observations of sleep duration. Green points represent days when exercise was completed, and red points represent days when exercise was not completed. Data source: Daily habits tracking dataset collected by Elena Donelson. Data accessed 2026-05-28.

Figure 2. Relationship between amount of sleep (hours) and coffee consumption (cups) from April–May 2026. Points represent individual daily observations of sleep duration and coffee consumption. The x-axis shows the amount of sleep recorded each day, while the y-axis shows the number of cups of coffee consumed. Data source: Daily habits tracking dataset collected by Elena Donelson. Data accessed 2026-05-28.

3.

a.

An affective visualization of my personal data could represent each day as a growing plant. The height of the stem could show how many hours I slept, the number of leaves could represent cups of coffee consumed, flowers could represent exercise, and roots could represent hours spent on schoolwork. Days with more sleep and exercise would appear healthier and more vibrant, while days with less sleep and more coffee would look weaker. This would create a visual representation of how my daily habits contribute to my overall well-being.

b.

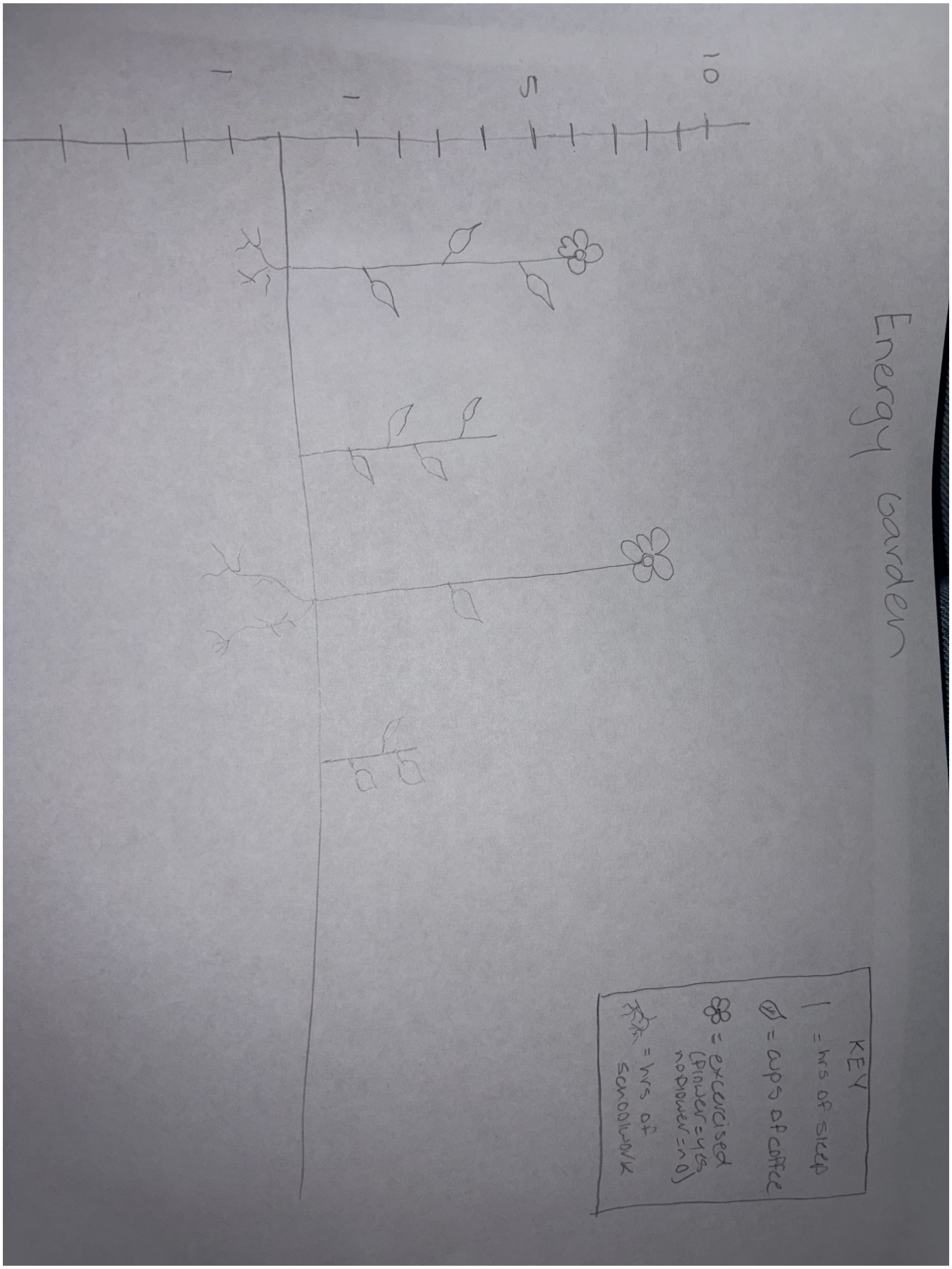


Figure 1: Sketch of affective visualization